

FFAR Annual Progress Report

As part of the Grant Agreement, grantees must complete an Annual Progress Report. Please use the template below to complete the programmatic report. The requirement must be submitted to FFAR within 30 days after the end of each annual funding period. All questions about this form should be directed to grants@foundationfar.org.

The Annual Progress Report communicates the annual results and accomplishments of the funded grant research, including accomplishments and tentative completion of specific annual goals and objectives. Disbursement of next year's funds for this grant are contingent on the receipt and approval of the Annual Progress Report to include a programmatic and financial piece as well as availability of matching funds, if applicable.

Grant Information	
Grant ID	430834
Award Program	New Innovator Award
Project Title	Building soil health through rotational diversity and soil microorganisms
Reporting Period	11/01/2020 – 10/31/2021
Period Budget	44,125.88 FFAR and 43,824.09 match
Project Director/Principal Investigator Information	
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1. General Information

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the U.S., please list the city and country.

This work is being conducted at several experimental research stations and on-farm across the state of Michigan. This includes the Agronomy Farm at MSU in Ingham County; Montcalm Research Center in Montcalm County; Saginaw Valley Research and Education Center in Saginaw county and; Kellogg Biological Station in Kalamazoo County. These research sites are located in the 3rd, 4th, 5th and 8th Michigan congressional districts.

1.2. How many new jobs were created by the grant during this reporting period?

No new jobs were created during this period. Delays in gaining no cost extension approval from FFAR led to expiration date in funding and therefore no funds were used.

1.3. How many jobs were maintained by the grant during this reporting period?

No jobs were maintained during this reporting period. Delays in gaining no cost extension approval from FFAR led to expiration date in funding and therefore no funds were used.

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

NO

2. Accomplishments

2.1. What were the goals/specific aims of the project for this reporting period? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 500-word limit)

The following research aims were initiated in the first year of the project and have not changed to this point.

Research Aim 1: Elucidate crop diversity effects on microbial community structure and diversity.

Research Aim 2: Increase understanding of microbial function, decomposition and N release between harvest and planting

Research Aim 3: Increase understanding of the interactions between rotational diversity and the potential for agroecosystem resilience in the face of winter climate change

2.2. Have any of the major goals/specific aims or milestones for this reporting period changed since the award or previous report? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 300-word limit)

No, there have been no changes in goals or project aims.

2.3. What was accomplished under the goals/specific aims or milestones for this reporting period? For this reporting period describe: 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include a discussion of stated goals not met. As the project progresses, the emphasis in reporting in this section should shift from reporting activities to reporting accomplishments. In the response, emphasize the significance of the findings to the scientific field. Include approaches taken to ensure robust and unbiased results. (up to 1,500-word limit)

During this reporting period there were advances in research related to this project with two graduate students working towards their degrees using data from previous years or funding from other sources for support. The primary focus of this research fell under research aims 2 and 3.

First, a Master's student just defended his thesis (August 2021) with research focused on determining the effects of increasing system diversity through addition of multiple cover crop species. More specifically, we are trying to understand the effects of functional, rather than species diversity by focusing on potential synergistic interactions between legume and grass cover crops. We measured microbial community activity and N availability through the growing season and throughout the winter in this experiment at the MSU Montcalm Research Center. The experimental plots were first established in September 2015 with soil samples collected approximately bi-monthly starting in September 2016 through October 2020. Collected soils were used to assess soil bulk density, soil moisture, water stable aggregates, N availability, soil respiration rates, extracellular enzyme activities (EEA), and microbial biomass C and N. We are also measuring above- and below-ground cover crop biomass and total soil C and N. Thus far, we have found greater aboveground biomass in single compared to mixed cover crop species, but mixed species produced the highest root biomass. The highest potential enzyme activities occur over winter (e.g. January) and were greatest in the two cover crop species mixes compared to four species mixtures. Total soil C and N (including particulate organic matter) was also greatest in the two cover crop species treatments compared to monocultures, four species and no cover crops.

By using several metrics to quantify microbial activity and N availability throughout the year we are beginning to better understand the timing of the break-down of different cover crop residues, which controls the timing of nutrient release and availability for potato uptake. Once published, our findings will be applicable to all Michigan field crop production as cover crops are a soil conservation tool growing in popularity by 2% per year (NCR-SARE

2014). By highlighting the positive and almost immediate impact of cover crops, this research can have immediate value to the potato industry (over 44,000 acres and \$166 million farm gate value in Michigan), as potato yield potential is markedly influenced by soil quality and N-availability. As biomass from cover crops is decomposed and converted to plant available N by soil microbes, N application in a given crop cycle can be reduced. For example, if all the potato production acres were planted to legume cover crops that fix 100 lb N/acre (Wilke and Snapp, 2008), \$0.58/lb for 28% liquid N, the value of the fixed N for the subsequent crop would be approximately \$2.3 million.

A second field study, supported by the USDA NIFA, is exploring the effects of early interseeding of cover crops with corn. We have completed this study and a PhD student defended his dissertation, including results from this project, in April 2022. The use of interseeded cover crops may be an ideal method for improving N use efficiency to help maximize productivity and sustainability of row crop agriculture. In the current study we are interseeding annual ryegrass, crimson clover or a mix of the two cover crops into V6 corn. We are assessing the dynamics of N cycling and provisioning with interseeding by measuring corn tissue N as well as soil inorganic N availability, microbial biomass N, peptidase extracellular enzyme activities, and nitrification and denitrification potentials. We have measured the soil parameters 11 times, starting after cover crops established in 2017 until 2020. The primary findings from this project were that early interseeded cover crops do not impact corn yields and in fact, can increase corn tissue N content. From a soil health and N cycling perspective, we found that location – i.e. soil edaphic factors – were the biggest drivers of soil responses to interseeded cover crops. We found that after two years of cover cropping, cover-cropping treatments had no effect on available C and N or on microbial nutrient-cycling activities in most site years. We observed major differences between sites, including in associations with soil nutrient pools and microbial activities. Differences in the availability of dissolved organic C (DOC) appeared to be the primary driver of N cycling at the two locations. Despite a more rapid flux of N through the N cycle, one location with finer-textured soils and qualitatively better soil health exhibited lower potential for N loss.

Finally, in this reporting period, we also assessed the potential for N loss via denitrification in the corn rhizosphere under monoculture and interseeded cover crop treatments. The rhizosphere is a hotspot of microbial activity, where the chemical composition and quantity of C is quite variable depending on both plant species and plant community diversity. However, we lack an understanding of how rhizosphere-driven differences in C will impact denitrification. We assessed denitrification in laboratory assays of bulk soils and rhizosphere soils of monoculture maize (*Zea mays* L.) and maize interseeded with annual ryegrass (*Lolium multiflorum* Lam.) and crimson clover (*Trifolium incarnatum* L.). In addition, we measured potential denitrification in rhizosphere soils of ryegrass and clover. We found that denitrification potential was enhanced in all rhizosphere soils compared to bulk soil and this effect varied depending on the species of plant, with significantly increased rates of denitrification in maize rhizosphere compared to clover rhizosphere. The availability of dissolved organic C, but not microbial biomass C, accounted for differences in denitrification rates among rhizosphere soils from different plants. Although not consistently enhanced in the rhizosphere, the reduction of N₂O in bulk soils was significantly lower when maize was intercropped with ryegrass compared to maize grown with clover or in monoculture, demonstrating the effects of increased plant diversity on denitrifiers. This study indicates how denitrification rates can differ significantly within the rhizospheres of different plants and highlights the importance of C availability as a driver of differences in denitrification rates in the rhizospheres of maize and cover crops.

2.3.1. Aside from the accomplishments outlined above, have there been any other significant impacts resulting from the work under this grant? Please describe any broader impacts such as:

- *Tools developed*



- *Benefits to policy*
- *Benefits to future research*
- *Benefits to food, the food system or agriculture*
- *Broader economic or health benefits*

If there are no broader impacts to date, state "Nothing to Report."

Nothing to report.

2.4. Describe challenges or delays encountered during the reporting period and actions or plans to resolve them. Only describe significant challenges that may impede the research and emphasize their resolution. (up to 500-word limit)

Communication delays between FFAR accounting and MSU Contracts and Grants delayed the approval of a no cost extension to the grant such that remaining funds from FFAR were never available in this reporting period.

2.5. Have there been any changes in scientific approach or reasons for change? If so, what are the changes? Remember, changes to the approved scientific approach must be pre-approved by FFAR. (up to 500-word limit)

No changes.

2.6. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Two graduate students worked on this project during the reporting period, writing either a thesis or dissertation using data collected through support from this project. Their training included gaining experience with scientific writing, learning new statistical analysis methods and organizational skills.

2.7. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 300-word limit)

Two graduate students worked on this project during the reporting period.

2.8. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in



learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 500-word limit)

We have reported results from the various studies described above to national and local scientific and stakeholder audiences. We have reached other MSU students and educators with public thesis and dissertation defenses. Data have been presented to corn and potato growers and the Michigan Agribusiness Association. Nationally, results have been shared with researchers and growers at Soil Ecology Society and American Geophysical Union meetings.

2.9. What do you plan to do during the next reporting period to accomplish the goals of the approved project? Briefly describe what you plan to do during the next reporting period to accomplish the project goals and objectives. Discuss efforts to ensure the approach is scientifically rigorous and results are robust and unbiased. Remember that significant changes to the approved goals and objectives and project scope require prior FFAR approval. Include any important modification to the original goals and provide justification for the change. (up to 500-word limit)

No new work will be completed in the next period. Another no cost extension will be requested so that the remaining funds can be used to pay salary and expenses incurred in this reporting period for this project, but were charged to other projects in my lab.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models

software, curricula, instruments or equipment, intervention, etc.) produced during this reporting period resulting directly from the FFAR award.

Dissertation was completed, thesis still in progress. Chapters from the dissertation are being submitted to journals for publication.

3.2. Please provide a list of citations for the information products produced during this period.

Curtright, A. J. (2022). Carbon-Mediated Ecological and Physiological Controls on Nitrogen Cycling across Agricultural Landscapes (Doctoral dissertation, Michigan State University).

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the reporting period resulting directly from the FFAR award? If yes, please provide citation.

N/A

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

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3.5. Have inventions, patent applications, and/or licenses resulted from the award during this reporting period? If yes, indicate the invention, patent application(s), and/or license(s).

N/A

3.6. Are any of the information products produced during this reporting period confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

N/A

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily

discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

Data sets will be submitted to Dryad once papers are accepted for publication.

4. Data Management

4.1. During this period, did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated in this reporting period.

Data generated includes basic biogeochemical assay data, soil and site characterization data including soil respiration rates, extracellular enzyme activities (EEA), inorganic N, microbial biomass C and N, water stable aggregates, bulk density, above- and below-ground cover crop biomass and total soil C and N.

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

These data sets are related in that they are measured at the same sites for the same projects.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Upload copies of records and a simple file inventory, if necessary, in a compressed folder.

N/A

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

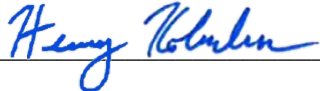
If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.



PI Full Name: Lisa Kay Tiemann

PI Signature:  Date: 9/8/22

Authorized Signing Official (ASO): Henry Kokenakes

ASO Signature:  Date: 9/8/2022